Client Meeting #2 Notes

Raw Data & Organization

- No strict requirements for raw data format → main priority is correctness.
- Data can be organized however makes the most sense for workflow.
- Most restrictive datasets: H3K27ac enhancers across species.
- Hardest species for data: pig and cat.
- Possible strategy: expand species list (papers show ~20 liver species available).

Enhancers & Distance to TSS

- Enhancers can be far from their target genes.
- Current search window capped at 1000 bp is too restrictive; acceptable to increase distance.
- Standard practice: up to 1 Mb (1,000,000 bp).
- More human data is fine—humans may have 40+ tissues.

CTCF & Tissue Context

- For CTCF/3D genome, need same tissue context across species.
- Likely will need to limit to one tissue (or a few tissues).
- Goal: Use CTCF to define regulatory domains and connect enhancers to genes.

Analysis Goals

1. Define enhancer groups/labels (e.g., conserved vs. human-specific).
2. Associate enhancer groups with outcomes:
    - RNA expression levels.
    - GWAS hits in enhancer regions.
    - CTCF binding strength & conservation.
3. Long-term: Partitioned heritability analysis
    - Test whether conserved enhancers/CTCF regions are enriched for trait-associated variants.

Web App Development

- Current focus: visualization & characterization of enhancers.
- Future: build in association tests/heritable analysis directly in the app.
- Potential workflow: upload GWAS hits → app runs partitioned heritability → output results with visualizations.

Server & Deployment

- For class project, downtime not a big issue.
- No need for complex rolling deployments.
- If ever needed: updates could be scheduled during off-hours (U.S. business hours most active).

Grading / Person Days

- “Person days” are approximate → some subjectivity.
- Target ~10 person days per sprint, but flexibility is allowed:
  - Could be fewer high-impact tasks.
  - Could be more small/administrative tasks.
- Descriptions and reflections more important than exact numbers.
- Reflections should capture challenges (e.g., trying multiple failed approaches before success).

Next Steps

- Continue enhancer annotation across species.
- Start CTCF analysis tab in the app.
- Focus on defining enhancer-gene connections by distance and CTCF domains.
- Move toward hypothesis testing after characterizing enhancer groups.